Ritik Vaishnav

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Education

- **MBM University** — Jodhpur, India
  Bachelor of Engineering - Electronics and Electrical Engineering — *July 2021 - June 2026*

Experience

- **Wireless Systems Engineering Lab - IIIT Delhi** — Delhi, India
  Research Intern under Dr. Arani Bhattacharya — *June 2025 - October 2025*
  - Designed and implemented a stereo-guided adaptive super-resolution pipeline achieving 35% improvement in far-object detection (>100m) for autonomous vehicles.
  - Implemented StereoSGBM-based disparity estimation and developed depth-based region partitioning to identify far-field regions for selective super-resolution.
  - Developed a depth-aware selective tiling framework that reduces SR computation by 3× compared to full-frame processing.
  - Optimized the end-to-end Stereo-SR pipeline on GPU (RTX 3060) for real-time performance (∼80 ms/frame).
  - Validated the framework on CARLA and KITTI, demonstrating 27% improvement in 50–100m detection with depth estimation MAE of 5.70m.
- **Wireless Systems Engineering Lab - IIIT Delhi** — Delhi, India
  Research Intern under Dr. Arani Bhattacharya — *June 2024 - July 2024*
  - Implemented efficient video encoding and streaming pipeline using FFmpeg (H.264/H.265) and RTSP in CARLA, achieving >80% bandwidth reduction with PSNR >30 dB.
  - Enabled real-time remote teleoperation through keyboard-based vehicle control integration in CARLA simulator.
  - Optimized Pylot perception pipeline by replacing Faster-RCNN with YOLOv8n, enabling dynamic cloud-based model switching to YOLOv8x for complex scenarios.
- **Embedded Systems & Robotics Workshop** — Jodhpur, India
  Robotics Mentor — *June 2023 - August 2023*
  - Mentored a batch of 100+ undergraduates in electronics and programming, sharing knowledge gained from ESRC under Prof. Alok Singh Gahlot.

Publications & Recognition

- **How Far Is Too Far? Fixing Vision of Autonomous Vehicles Using Selective Super-Resolution [Paper]**
  Najiya Naj, Ritik Vaishnav, and Arani Bhattacharya — [COMSNETS-W]
  Workshop on Machine Intelligence in Networked Data and Systems — Jan 2026
- **Best Presentation Award** — [RIISE]
  Poster: “How Far is Too Far? Fixing AV vision with the cloud” — Sept 2025
  RIISE 2025, IIIT Delhi

Projects

- **StereoSR for Autonomous Vehicle Perception** — Sept 2025
  - Implemented stereo vision pipeline for depth estimation using dual camera setup in CARLA simulator.
  - Achieved strong accuracy with ground truth validation against CARLA’s native depth sensor.
  - Benchmarked the setup on KITTI Stereo dataset with an MAE of 5.70m, improving far-object detection by 35% beyond 100m.
- **PaperPerspective** — April 2024
  - Implemented perspective transformation in real-time, eliminating the need for camera or whiteboard setups.
  - Utilized OpenCV and pyvirtualcam for seamless virtual camera integration.
- **Autonomous Rover with Real-Time Video Feedback** — July 2022
  - Integrated an ESP32 Camera with a Raspberry Pi Pico-driven fast line-follower rover.
  - Ensured seamless real-time video streaming through a Live Web Server.
  - Optimized PID control parameters for enhanced accuracy and smooth trajectory following.

Position of Responsibility

- **Founder, Pixellens: Film and Photography Club** Jodhpur, India
  Founded and built the club from ground up *March 2022 - March 2025*
  - Grew it to a community of 50+ super creative people
  - Created and led multiple creative film and photography projects
  - Written/Directed/Shot/Edited short films in 50Hr timeline - 3x

Skills

- **Languages**: Python, C++
- **Frameworks**: PyTorch, TensorFlow, OpenCV, YOLO
- **Tools**: ROS, FFmpeg, CARLA, Pylot, Git, Linux

Interests

- **Research Interests**: Autonomous Vehicle Perception, Edge Computing, Real-time Systems, Computer Vision, Hardware-Software Co-design
- **Abstract Interests**: Filmmaking, Music